

Extra Exercise #3

#1. V is constant ; $Q=0$

state 1 Ammonia

$$V_{\text{sat. liquid}} = 0.005 \text{ m}^3 (V_f)$$

$$V_{\text{sat. vapor}} = 1.5 \text{ m}^3 (V_g)$$

$$P_1 = 5 \text{ bar} ; v_f = 0.00158 \text{ m}^3/\text{kg}$$

state 2

$$P_2 = ? ; P_2 > P_1$$

$$V_{\text{sat. liquid}} = 0 ; V_{\text{sat. vapor}} = ?$$

$$v_g = 0.2503 \text{ m}^3/\text{kg}$$

$$v_f = \frac{V_f}{m_f} ; m_f = \frac{V_f}{v_f}$$

$$m_f = 0.005 \text{ m}^3 / 0.00158 \text{ m}^3/\text{kg} = 3.165 \text{ kg}$$

$$v_g = \frac{V_g}{m_g} ; m_g = \frac{V_g}{v_g} = \frac{1.5 \text{ m}^3}{0.2503 \text{ m}^3/\text{kg}} = 5.993 \text{ kg}$$

$$x = \frac{m_g}{m_f + m_g} = \frac{5.993 \text{ kg}}{3.165 \text{ kg} + 5.993 \text{ kg}} = 0.6544$$

$$v_1 = v_f + x (v_g - v_f) ; v_1 = 0.00158 + 0.6544 (0.2503 - 0.00158)$$

$$v_1 = 0.1643 \text{ m}^3/\text{kg} ; v_1 = v_2$$

$$v_2 = 0.1643 \text{ m}^3/\text{kg} = v_g$$

from tables

Method of Interpolation

$$\frac{P_1 - P_2}{P_1 - P_x} = \frac{v_1 - v_2}{v_1 - v_x} ; \frac{7 - 8}{7 - P_x} = \frac{0.1815 - 0.1596}{0.1815 - 0.1643}$$

$$\frac{-1}{7 - P_x} = 1.2733 ; -1 = 8.913 - 1.2733 P_x$$

$$\textcircled{a} P_x = 7.7853 \text{ bar} ;$$

$$m = m_f + m_g = 3.165 + 5.993 = 9.158 \text{ kg}$$

$$\cancel{\Delta P \cdot E} + \cancel{\Delta K \cdot E} + \Delta u = \cancel{Q} + W \Rightarrow \Delta u = W$$

$$(b) \quad u_{if} = 198.39 \frac{\text{kJ}}{\text{kg}} \quad u_{ig} = 1321.02 \frac{\text{kJ}}{\text{kg}}$$

$$u_1 = u_{if} + x (u_{ig} - u_{if}) \quad ; \quad u_1 = 198.39 + 0.6544(1321.02 - 198.39)$$

$$u_1 = 933.04 \frac{\text{kJ}}{\text{kg}}$$

$u_2 \Rightarrow$ Method of Interpolation

$$\frac{v_1 - v_2}{v_1 - v_x} = \frac{u_1 - u_2}{u_1 - u_x} \Rightarrow \frac{0.1815 - 0.1596}{0.1815 - 0.1643} = \frac{1328.04 - 1330.64}{1328.04 - u_x}$$

$$1.2733 = \frac{-2.6}{1328.04 - u_x} \quad ; \quad 1690.99 - 1.2733 u_x = -2.6$$

$$u_x = 1330.1 \frac{\text{kJ}}{\text{kg}} = u_2$$

$$; \quad (u_2 - u_1)m = W$$

$$(1330.1 - 933.04)9.158 = W$$

$$W = 3636.3 \text{ kJ}$$

#2 $m = 18 \text{ kg}$ Ammonia

state 1

$$P_1 = 100 \text{ kPa}; T_1 = -10^\circ\text{C}; u_1 = 1324.33 \frac{\text{kJ}}{\text{kg}}$$

from tables $v_1 = 1.2621 \frac{\text{m}^3}{\text{kg}}$

state 2

$$P_2 = 550 \text{ kPa}$$

$$n = 1.1$$

$$P_1 v_1^{1.1} = P_2 v_2^{1.1}; 100 \text{ kPa} \times 1.2621^{1.1} = 550 \text{ kPa} v_2^{1.1}$$

$$v_2^{1.1} = 0.23488; v_2 = 0.26794 \frac{\text{m}^3}{\text{kg}}$$

$$\textcircled{a} W = \frac{P_2 v_2 - P_1 v_1}{1-n}; m \frac{(P_2 v_2 - P_1 v_1)}{1-n}$$

$$= \frac{18 \text{ kg} (550 \times 0.26794 - 100 \times 1.2621) \text{ kPa} \frac{\text{m}^3}{\text{kg}}}{1-1.1} = -3808.26 \text{ kJ}$$

$$\textcircled{b} \Delta P.E + \Delta K.E + \Delta u = Q + W$$

$$\text{for State 2} \quad \frac{v_1 - v_2}{v_1 - v_x} = \frac{u_1 - u_2}{u_1 - u_x}$$

$$\frac{0.2644 - 0.27454}{0.2644 - 0.26794} = \frac{1389.64 - 1408.53}{1389.64 - u_x}; 1389.64 - u_x = -6.5947$$

$$u_x = 1396.2 \frac{\text{kJ}}{\text{kg}} = u_2$$

$$Q = \Delta u - W; [18 \text{ kg} (1396.2 - 1324.33) \frac{\text{kJ}}{\text{kg}}] + 3808.26 \text{ kJ}$$

$$Q = 5101.92 \text{ kJ}$$

#3, $m = 0.25 \text{ kg}$

$T_1 = 200^\circ\text{C} + 273.15 = 473.15 \text{ K}$

$P_1 = 2 \text{ bar}$

$M = 28.97 \frac{\text{kg}}{\text{kmol}}$ $P_2 = 2 \text{ bar}$ $V_2 = 2V_1$

$R_{\text{air}} = \frac{\bar{R}}{M} = \frac{8.314 \text{ kJ}}{\text{kmol} \cdot \text{K}} \times \frac{1 \text{ kmol}}{28.97 \text{ kg}} = 0.287 \frac{\text{J}}{\text{g} \cdot \text{K}}$

$V_1 = \frac{mRT}{P} = 0.25 \text{ kg} \times 0.287 \frac{\text{J}}{\text{g} \cdot \text{K}} \times 473.15 \text{ K} \times \frac{1}{2 \text{ bar}} \times \frac{1 \text{ bar}}{102 \text{ kPa}} = 0.1697 \text{ m}^3$

$V_2 = 2V_1 = 0.3394 \text{ m}^3$

$W = -\int P \Delta V = 2 \text{ bar} \times \frac{10^2 \text{ kPa}}{1 \text{ bar}} \times (0.3394 - 0.1697) \text{ m}^3 = -33.94 \text{ kJ}$

$33.94 \text{ kJ} = 33940 \text{ J} \times \frac{1 \text{ Btu}}{1055.06 \text{ J}} = -32.17 \text{ Btu}$

© $\frac{T_2}{T_1} = \frac{V_2}{V_1}$; $\frac{T_2}{473.15} = \frac{0.3394}{0.1697}$; $T_2 = 946.3 \text{ K}$

$Q = mC_p(T_2 - T_1)$ $K = 1.4$ $C_p = 1.005$

$Q = 0.25 \text{ kg} \times 1.005 (946.3 - 473.15)$

$Q = 118.88 \text{ kJ}$

#4 Air $M = 28.97 \frac{\text{kg}}{\text{kmol}}$

$P_1 = 0.7 \text{ bar}$ $T_1 = 275 \text{ K}$ $V_1 = 0.25 \text{ m}^3$ $V_2 = 0.03 \text{ m}^3$

$PV^{1.25} = \text{Constant}$ $k = ?$ $Q = ?$ $n = 1.25$

$P_1 V_1^{1.25} = P_2 V_2^{1.25}$; $P_2 = \frac{P_1 V_1^{1.25}}{V_2^{1.25}} = \frac{0.7 \text{ bar} \times 0.25^{1.25}}{0.03^{1.25}}$

$P_2 = 9.911 \text{ bar}$

$W = \frac{P_2 V_2 - P_1 V_1}{1 - n} = \frac{(9.911 \times 0.03) - (0.7 \times 0.25)}{1 - 1.25} = -0.48932 \text{ kJ}$

$R = \frac{\bar{R}}{M} = 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \times \frac{1 \text{ kmol}}{28.97 \text{ kg}} = 0.287 \frac{\text{J}}{\text{K} \cdot \text{g}}$

$PV = mRT$; $m = \frac{P_1 V_1}{RT_1}$

$m = 0.7 \text{ bar} \times \frac{10^2 \text{ kPa}}{1 \text{ bar}} \times 0.25 \text{ m}^3 \times \frac{1 \text{ K} \cdot \text{g}}{0.287 \text{ J}} \times \frac{1}{275 \text{ K}} = 0.2217 \text{ kg}$

$n = \frac{m}{M} = 0.2217 \text{ kg} \times \frac{1 \text{ kmol}}{28.97 \text{ kg}} = 0.00765 \text{ kmol}$

$C_v = \frac{5}{2} \bar{R}$; $\Delta U = n C_v (T_2 - T_1)$

$= 0.00765 \text{ kmol} \times \frac{5}{2} \times 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \times \left(\frac{P_2 V_2}{n R} - 275 \right)$